

RA-IZ, Russian Federation

WMO: 233310

Lat: 66.852N

Lon: 65.299E

Elev: 890

StdP: 91.08

Time Zone: 5.00 (E05)

Period: 82-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.9	-30.4	-35.0	0.2	-32.1	-32.7	0.2	-29.4	34.2	-12.5	33.8	-12.1	8.2	0	1.112

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	19.2	14.4	16.7	12.3	14.8	11.2	15.1	18.3	13.3	15.7	11.7	14.0	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	10.8	16.3	12.1	9.8	15.0	10.5	8.8	13.0	45.1	18.4	40.2	15.6	35.9	14.0	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	24.0	19.6	DB	-36.3	21.9	4.2	4.1	-39.3	24.9	-41.7	27.3	-44.1	29.6	-47.1	32.6	(4)
(5)			WB	-35.6	16.9	3.5	2.8	-38.1	18.9	-40.1	20.6	-42.1	22.1	-44.6	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.1	-18.2	-17.8	-12.5	-10.8	-5.5	3.7	9.1	5.8	0.2	-7.8	-14.3	-17.7	(6)
(7)		DBStd	10.94	6.23	6.61	4.84	7.21	5.40	6.37	5.18	4.18	4.08	5.58	5.93	6.00	(7)
(8)		HDD10.0	6315	873	779	699	625	482	202	81	142	295	551	728	858	(8)
(9)		HDD18.3	9280	1131	1012	957	875	739	438	289	390	544	809	978	1117	(9)
(10)		CDD10.0	80	0	0	0	0	1	14	54	11	1	0	0	0	(10)
(11)		CDD18.3	3	0	0	0	0	0	0	3	0	0	0	0	0	(11)
(12)		CDH23.3	4	0	0	0	0	0	0	4	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.3	8.1	7.3	7.9	7.8	7.1	6.3	5.8	5.9	6.9	8.4	7.5	8.1	(14)
(15)	Precipitation	PrecAvg	875	56	49	68	70	72	79	79	86	72	105	75	64	(15)
(16)		PrecMax	1036	88	80	148	144	115	116	161	171	144	150	129	97	(16)
(17)		PrecMin	758	37	19	20	29	31	34	16	37	22	39	48	39	(17)
(18)		PrecStd	62	13	16	21	23	22	23	32	31	26	23	19	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.0	-4.5	-2.7	8.0	14.5	17.8	24.1	17.1	13.4	4.6	1.2	-4.4	(19)
(20)			MCWB	-6.1	-5.9	-4.1	7.4	9.2	12.6	16.8	11.7	11.4	2.6	0.0	-5.2	(20)
(21)		2%	DB	-6.5	-5.9	-4.1	2.8	8.0	16.0	21.6	14.7	9.4	2.9	-3.7	-5.9	(21)
(22)			MCWB	-7.3	-6.9	-4.9	1.1	5.7	11.1	16.2	11.0	7.5	1.9	-4.4	-6.6	(22)
(23)		5%	DB	-8.1	-7.4	-5.2	1.0	4.2	14.1	18.8	13.1	7.6	1.3	-5.0	-7.3	(23)
(24)			MCWB	-8.7	-8.2	-5.8	-0.7	2.4	10.1	14.2	10.0	6.5	0.7	-5.6	-7.9	(24)
(25)		10%	DB	-9.9	-9.3	-6.3	-0.8	1.8	12.3	16.7	11.5	5.9	-0.4	-6.1	-9.1	(25)
(26)			MCWB	-10.4	-10.0	-7.0	-2.1	0.2	8.6	12.5	9.4	4.9	-0.7	-6.5	-9.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.7	-5.1	-3.5	7.3	9.6	14.2	18.4	13.8	11.7	3.4	-0.7	-5.1	(27)
(28)			MCDWB	-5.3	-4.9	-3.1	8.0	13.2	16.4	22.2	15.5	13.5	3.9	1.3	-4.5	(28)
(29)		2%	WB	-7.2	-6.8	-4.8	1.4	5.7	12.1	16.4	12.1	8.0	1.9	-4.3	-6.4	(29)
(30)			MCDWB	-6.6	-6.2	-4.3	2.6	7.5	14.3	20.7	13.6	9.2	2.7	-3.9	-6.0	(30)
(31)		5%	WB	-8.7	-8.3	-5.7	-0.3	2.8	10.5	14.8	10.7	6.6	0.6	-5.5	-7.9	(31)
(32)			MCDWB	-8.0	-7.6	-5.1	0.8	4.2	13.6	18.1	12.1	7.3	1.3	-5.1	-7.3	(32)
(33)		10%	WB	-10.3	-9.9	-6.8	-2.2	0.4	9.2	13.4	9.6	5.1	-0.7	-6.6	-9.6	(33)
(34)			MCDWB	-9.9	-9.3	-6.4	-0.9	1.7	11.9	15.8	11.1	5.6	-0.3	-6.1	-9.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	5.6	4.6	5.8	5.2	5.8	5.2	4.3	3.4	3.8	4.5	5.2	(35)
(36)			MCDBR	7.2	7.4	4.7	5.9	8.0	7.3	6.7	6.3	5.6	6.1	4.8	5.9	(36)
(37)			MCWBR	6.8	6.8	4.3	4.9	7.0	5.3	4.9	4.6	5.5	4.8	4.5	5.6	(37)
(38)		5% WB	MCDWB	7.4	7.4	4.7	5.5	7.3	6.9	6.4	5.5	5.2	5.1	4.7	6.0	(38)
(39)			MCWBR	7.1	6.9	4.5	5.0	6.8	5.6	5.3	4.6	5.3	4.5	4.5	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.167	0.221	0.263	0.285	0.300	0.314	0.335	0.318	0.280	0.234	0.163	N/A	(40)	
(41)		taud	1.886	2.101	2.163	2.184	2.256	2.475	2.447	2.477	2.533	2.252	1.913	N/A	(41)	
(42)		Ebn at Noon	220	641	796	874	897	896	860	841	800	623	209	0	(42)	
(43)		Edh at Noon	14	50	82	106	111	93	93	81	59	42	12	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.06	0.55	2.07	4.03	5.31	4.96	5.06	3.18	1.72	0.56	0.12	0.00	(44)	
(45)		RadStd	0.01	0.07	0.23	0.32	0.45	0.44	0.65	0.29	0.16	0.09	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+1.54	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page